

# SH1017 Physics for Engineering Mathematics Lab 1

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August 17, 2026

## Q1: Period of Undamped Pendulum

**Method:** We calculate the period  $T$  of oscillation for different initial angular displacements  $\theta_0$  by simulating the differential equation  $\ddot{\theta} + \omega_0^2 \sin(\theta) = 0$  using the 4th order Runge-Kutta (RK4) method (`pendulum.py`). The simulation detects turning points to find the half-period. We compare this to a small angle approximation, a power series expansion, and direct numerical integration.

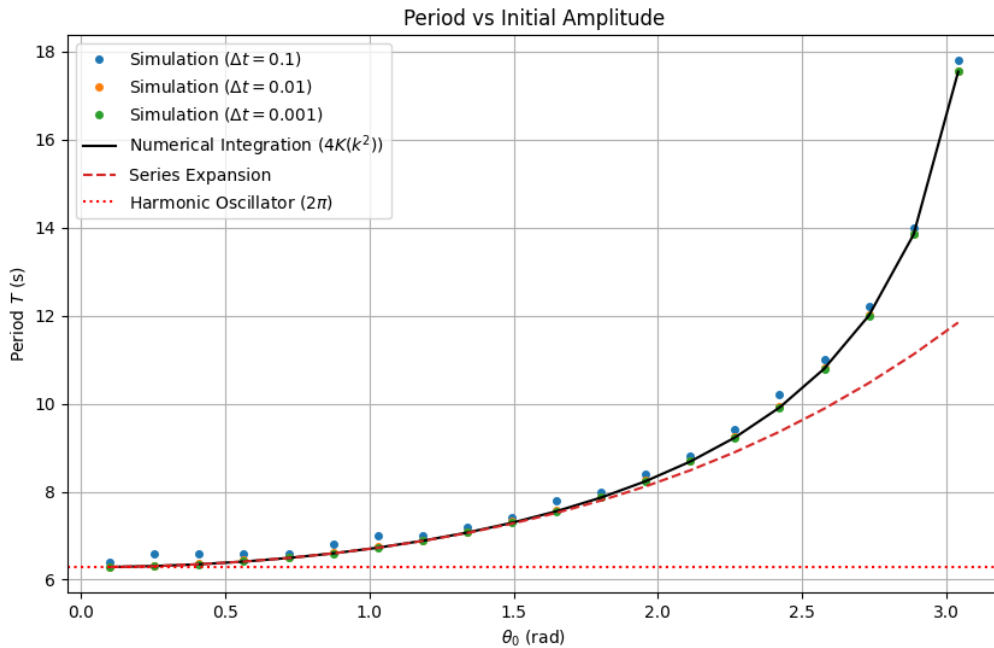


Figure 1: Period  $T$  as a function of the initial angular displacement  $\theta_0$ . Parameters:  $\theta_0 \in [0.1, \pi - 0.1]$  (20 linearly spaced values),  $\omega_0^2 = 1$ . Time steps:  $\Delta t \in \{0.1, 0.01, 0.001\}$ .

**Results:** We see in Figure 1 that all methods agree  $T \approx 2\pi$  for angles near zero. For larger angles, the period increases. The RK4 simulations for  $\Delta t = 0.1$  to  $\Delta t = 0.001$  coincides with the numerical integration. The series expansion gives an accurate approximation for angles below  $\theta_0 \approx \pi/2$  as the angle approaches  $\pi$ .

## Q2: Damped Pendulum

**Method:** We simulate the damped pendulum equation  $\ddot{\theta} + \gamma\dot{\theta} + \omega_0^2 \sin(\theta) = 0$  using RK4 (`pendulum.py`). Using the data from the simulation, we plot the natural logarithm of the positive peak amplitudes vs time (this information is used to find the  $\tau$  characteristic time).

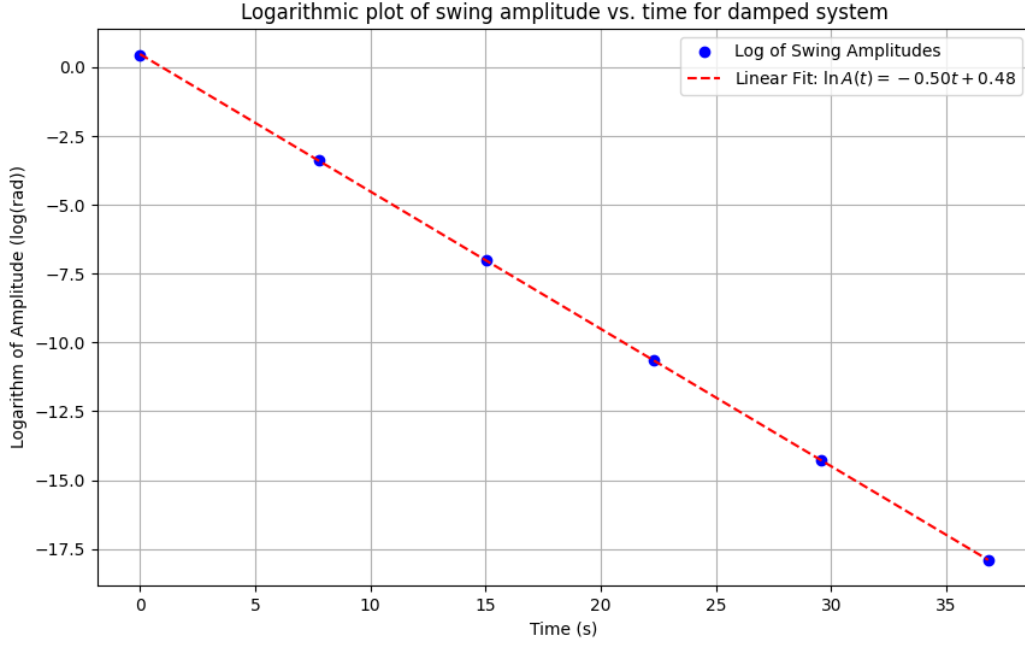


Figure 2: Logarithmic plot of swing amplitude vs. time for the damped system. Parameters:  $\theta_0 = \pi/2$ ,  $\gamma = 1.0$ ,  $\omega_0^2 = 1$ ,  $\Delta t = 0.01$ ,  $t_{\max} = 40$ .

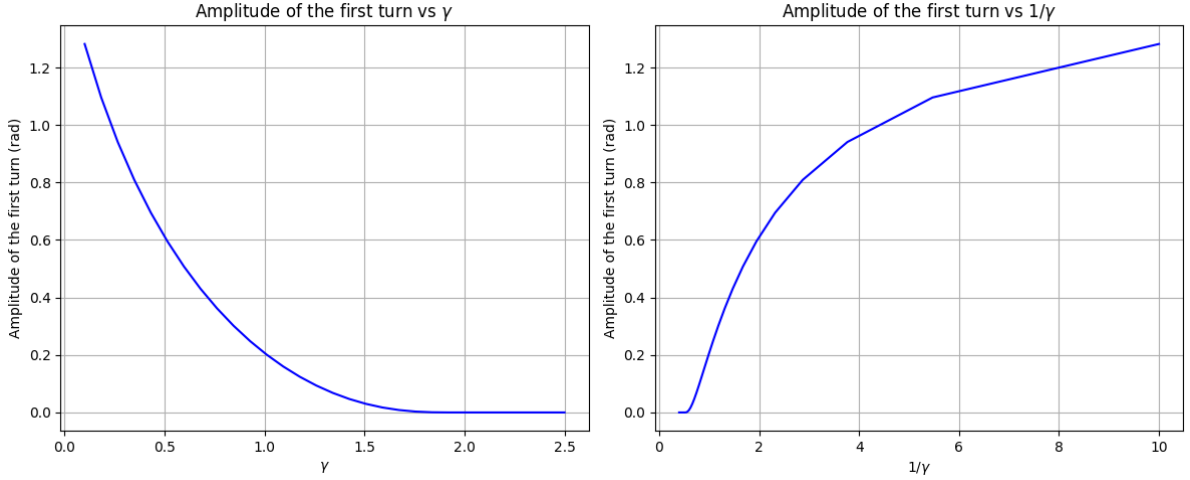


Figure 3: Comparison of the first turn amplitude for varying values of  $\gamma$  and  $1/\gamma$ . Parameters:  $\theta_0 = \pi/2$ ,  $\gamma \in [0.1, 2.5]$  (30 linearly spaced values),  $\omega_0^2 = 1$ ,  $\Delta t = 0.01$ ,  $t_{\max} = 20$ .

**Theory:** A weakly damped harmonic oscillator experiences an exponential decay in its amplitude over time,  $A(t) = A_0 e^{\lambda t}$ . This implies that  $\ln A(t)$  should decrease linearly with a slope of  $\lambda$ . This gives

$$\ln(\theta(t)) = \ln A_0 + \lambda t$$

To find the characteristic time  $\tau$  where the initial amplitude  $A_0$  is halved we first find  $\lambda$  through linear interpolation  $\lambda \approx \frac{\ln A_2 - \ln A_1}{t_2 - t_1}$  where  $A_2$  and  $A_1$  are distinct turning point amplitudes. Then we solve the equation

$$\ln A_0 + \lambda \tau = \ln\left(\frac{A_0}{2}\right) = \ln A_0 - \ln 2$$

$$\Longleftrightarrow \tau = -\frac{\ln 2}{\lambda}$$

Furthermore, as the damping coefficient  $\gamma$  increases, the system transitions toward an overdamped state. In the overdamped case, the pendulum should not oscillate or cross the equilibrium point, meaning its subsequent peak amplitude is zero.

**Results:** Figure 2 shows a strong linear relationship for  $\ln A(t)$  vs time, confirming that the nonlinear pendulum's envelope amplitude decays exponentially. The slope gives an estimated decay constant  $\lambda \approx -0.50$ , from which we calculate the characteristic time  $\tau = \ln(2)/\lambda \approx 1.39$  s. Figure 3 demonstrates that as  $\gamma$  increases, the amplitude of the first turn decreases. It reaches exactly zero at  $\gamma = 2.0$  (critical damping).

### Q3: Driven Damped Pendulum

**Method:** We simulate the driven damped system  $\ddot{\theta} + \gamma\dot{\theta} + \omega_0^2 \sin(\theta) = A \cos(\omega t)$  using RK4 (`pendulum.py`). We then measure the steady-state amplitude across various drive amplitudes  $A$ . We then generate a phase portrait ( $\theta$  vs  $p$ ) for a drive amplitude  $A = 1.0$  by discarding the initial transient segment.

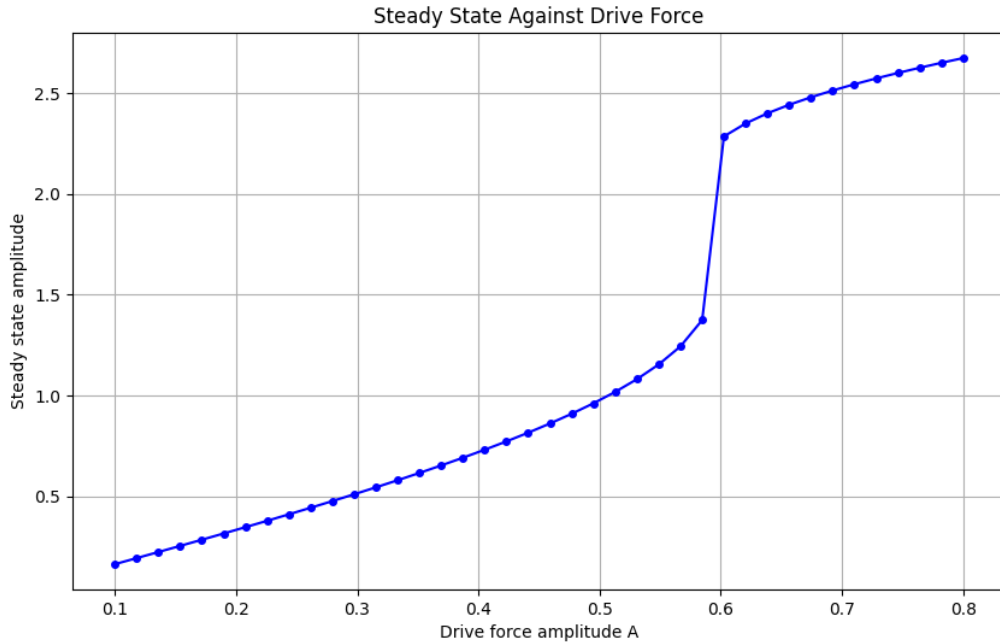


Figure 4: Steady state amplitude against drive force amplitude  $A$ . Parameters:  $\gamma = 3/8$ ,  $\omega = 2/3$ ,  $\omega_0^2 = 1$ ,  $A \in [0.1, 0.8]$  (40 linearly spaced values),  $\Delta t = 0.01$ , simulated until  $t = 1000$ .

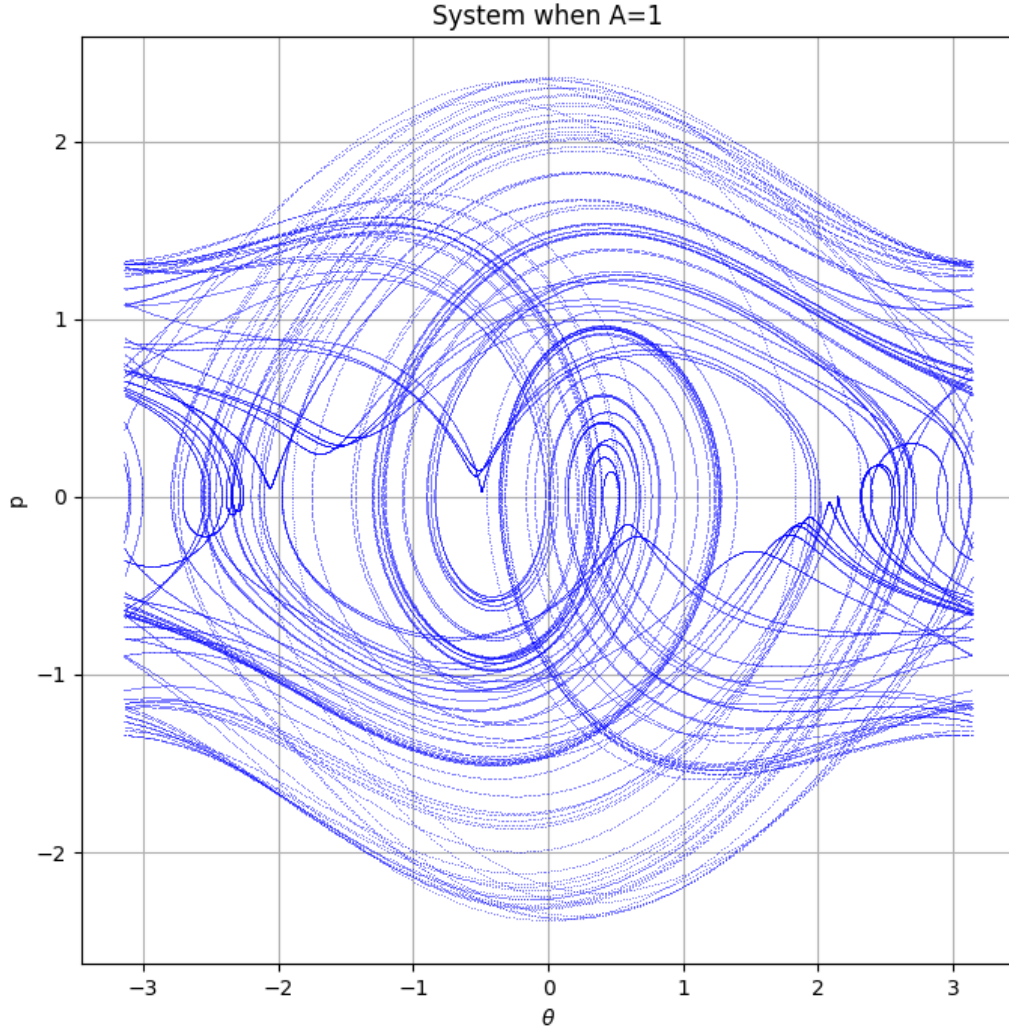


Figure 5: Phase portrait of the system when  $A = 1$ . Parameters:  $\gamma = 3/8$ ,  $\omega = 2/3$ ,  $\omega_0^2 = 1$ ,  $A = 1.0$ ,  $\Delta t = 0.01$ , simulated until  $t = 1000$ , transients discarded for  $t < 200$ .

**Results:** The steady state amplitude is obtained by observing when the amplitude of the pendulum's position converges. It seems to only reach a steady state with small enough values of the drive force amplitude  $A$ . If we imagine the turn amplitudes as a sequence namely  $X_1, X_2 \dots X_k$ , we stop at the amplitude such that  $|X_{k+1} - X_k| < \tau$  for a certain tolerance level  $\tau$ . (See figure 4). We choose tolerance  $\tau = 1e-6$ . When the drive force amplitude is  $1 \leq A$ , we find that the turn amplitude does not converge, i.e. the phase portrait does not reach a limit cycle. However, for  $A = 1$  the phase portrait does have periodic behavior as we can see in figure 5.